1. Description

1.1. Project

Project Name	Hephaestus
Board Name	custom
Generated with:	STM32CubeMX 6.15.0
Date	04/24/2026

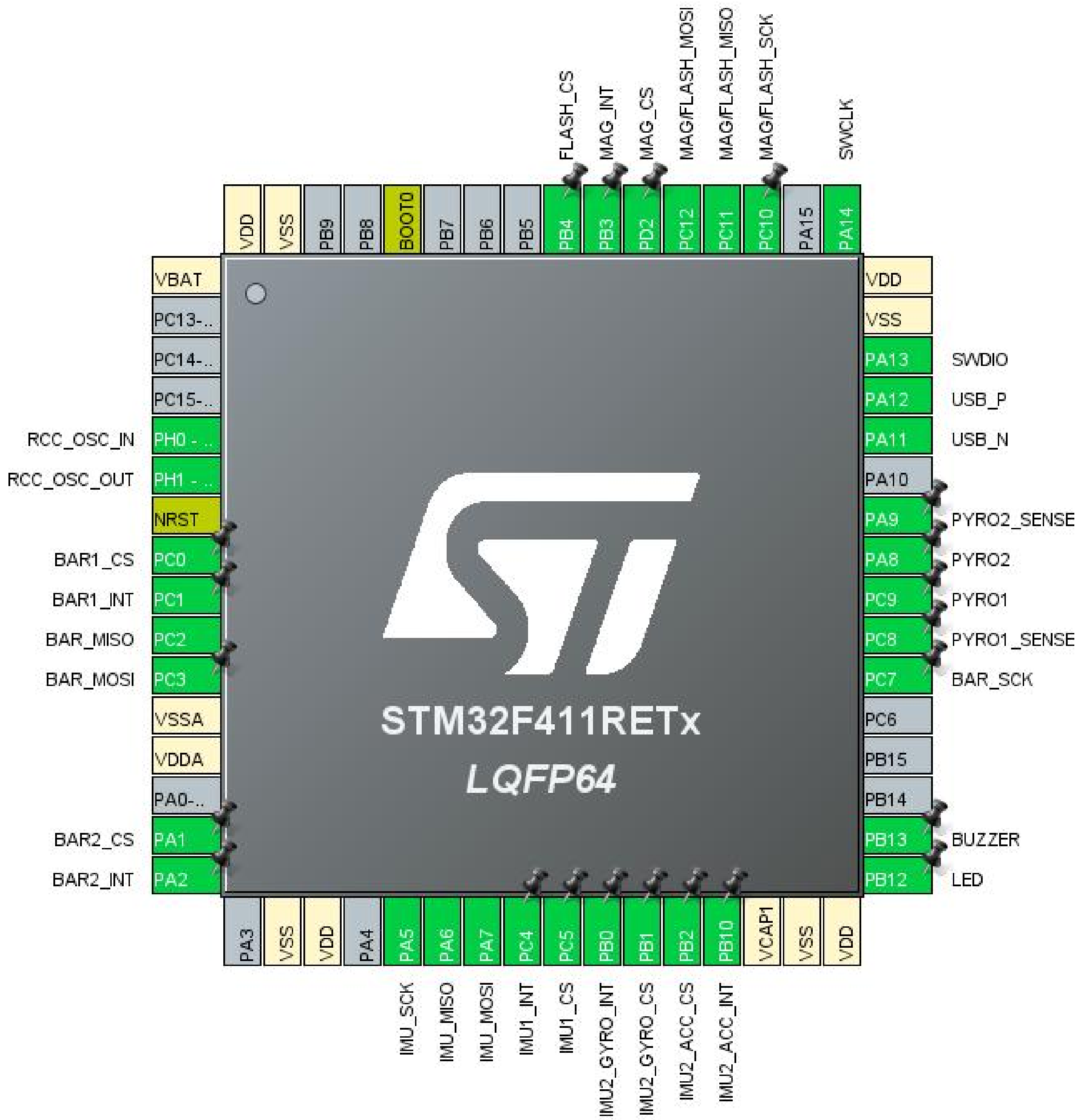
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F411
MCU name	STM32F411RETx
MCU Package	LQFP64
MCU Pin number	64

1.3. Core(s) information

Core(s)	Arm Cortex-M4

2. Pinout Configuration



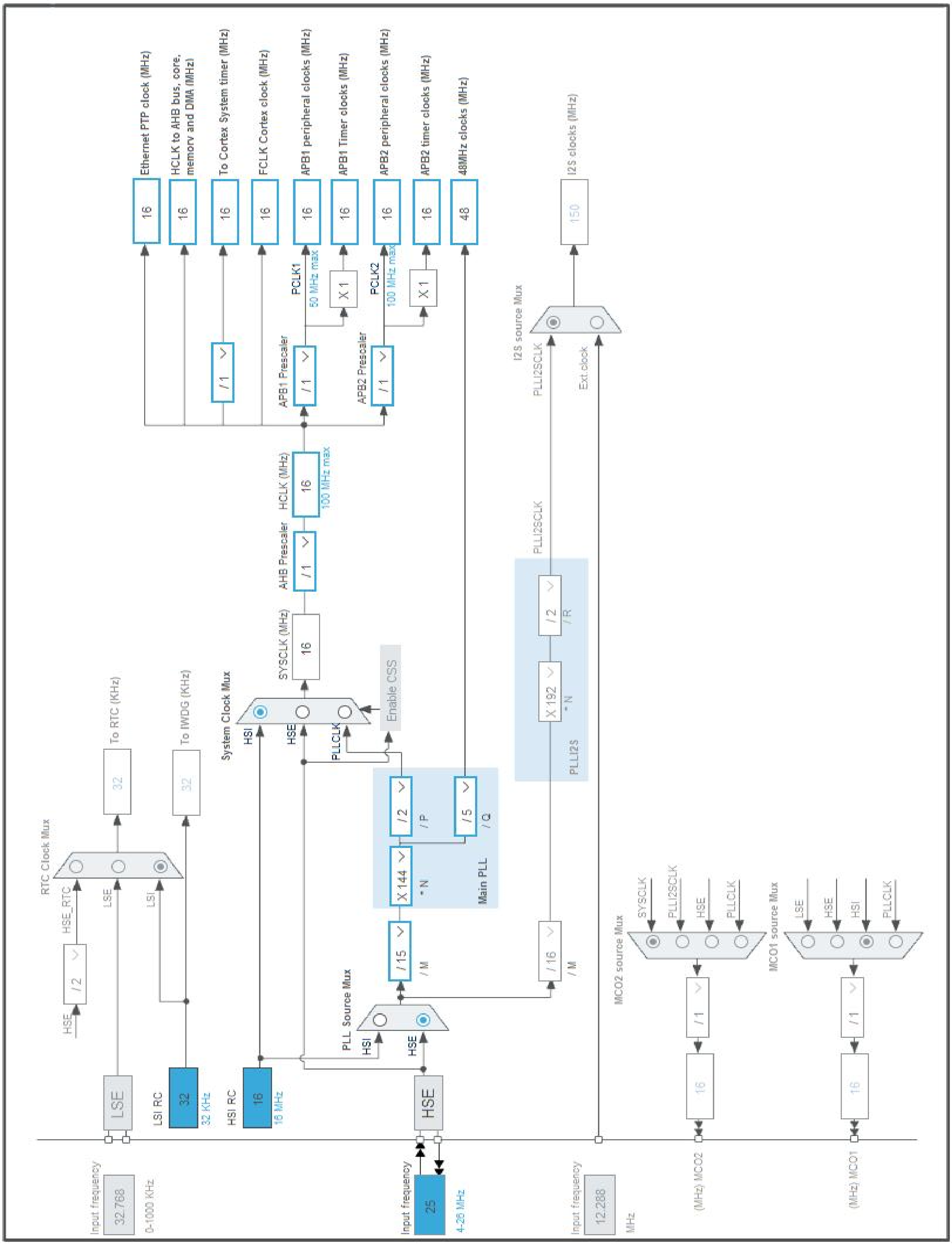
3. Pins Configuration

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	VBAT	Power		
5	PH0 - OSC_IN	I/O	RCC_OSC_IN	
6	PH1 - OSC_OUT	I/O	RCC_OSC_OUT	
7	NRST	Reset		
8	PC0 *	I/O	GPIO_Output	BAR1_CS
9	PC1	I/O	GPIO_EXTI1	BAR1_INT
10	PC2	I/O	SPI2_MISO	BAR_MISO
11	PC3	I/O	SPI2_MOSI	BAR_MOSI
12	VSSA	Power		
13	VDDA	Power		
15	PA1 *	I/O	GPIO_Output	BAR2_CS
16	PA2	I/O	GPIO_EXTI2	BAR2_INT
18	VSS	Power		
19	VDD	Power		
21	PA5	I/O	SPI1_SCK	IMU_SCK
22	PA6	I/O	SPI1_MISO	IMU_MISO
23	PA7	I/O	SPI1_MOSI	IMU_MOSI
24	PC4	I/O	GPIO_EXTI4	IMU1_INT
25	PC5 *	I/O	GPIO_Output	IMU1_CS
26	PB0	I/O	GPIO_EXTI0	IMU2_GYRO_INT
27	PB1 *	I/O	GPIO_Output	IMU2_GYRO_CS
28	PB2 *	I/O	GPIO_Output	IMU2_ACC_CS
29	PB10	I/O	GPIO_EXTI10	IMU2_ACC_INT
30	VCAP1	Power		
31	VSS	Power		
32	VDD	Power		
33	PB12 *	I/O	GPIO_Output	LED
34	PB13 *	I/O	GPIO_Output	BUZZER
38	PC7	I/O	SPI2_SCK	BAR_SCK
39	PC8 *	I/O	GPIO_Input	PYRO1_SENSE
40	PC9 *	I/O	GPIO_Output	PYRO1
41	PA8 *	I/O	GPIO_Output	PYRO2
42	PA9 *	I/O	GPIO_Input	PYRO2_SENSE
44	PA11	I/O	USB_OTG_FS_DM	USB_N
45	PA12	I/O	USB_OTG_FS_DP	USB_P
46	PA13	I/O	SYS_JTMS-SWDIO	SWDIO

Pin Number LQFP64	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
47	VSS	Power		
48	VDD	Power		
49	PA14	I/O	SYS_JTCK-SWCLK	SWCLK
51	PC10	I/O	SPI3_SCK	MAG/FLASH_SCK
52	PC11	I/O	SPI3_MISO	MAG/FLASH_MISO
53	PC12	I/O	SPI3_MOSI	MAG/FLASH_MOSI
54	PD2 *	I/O	GPIO_Output	MAG_CS
55	PB3	I/O	GPIO_EXTI3	MAG_INT
56	PB4 *	I/O	GPIO_Output	FLASH_CS
60	BOOT0	Boot		
63	VSS	Power		
64	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Power Consumption Calculator report

1.1. Microcontroller Selection

Series	STM32F4
Line	STM32F411
MCU	STM32F411RETx
Datasheet	DS10314_Rev6

1.2. Parameter Selection

Temperature	25
Vdd	1.7

1.3. Battery Selection

Battery	Li-SOCL2(A3400)
Capacity	3400.0 mAh
Self Discharge	0.08 %/month
Nominal Voltage	3.6 V
Max Cont Current	100.0 mA
Max Pulse Current	200.0 mA
Cells in series	1
Cells in parallel	1

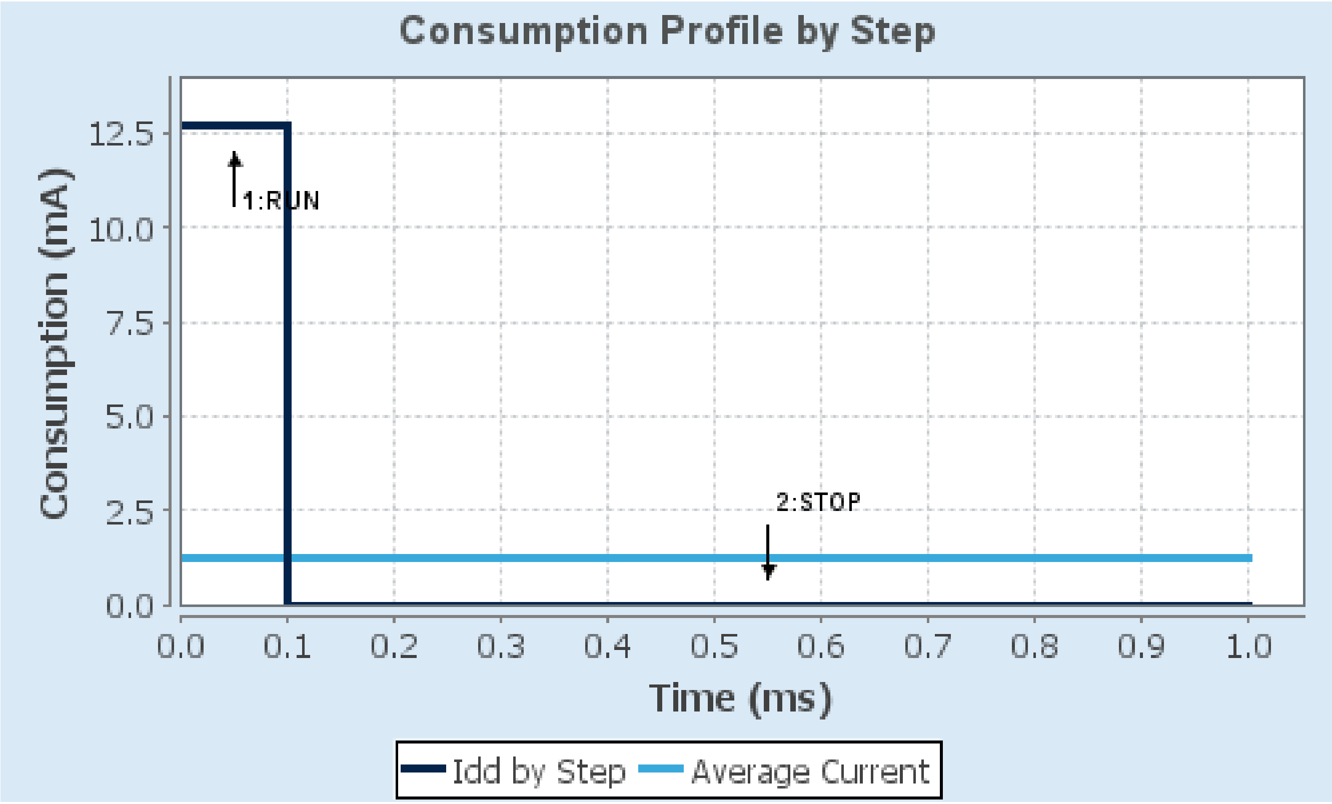
1.4. Sequence

Step	Step1	Step2
Mode	RUN	STOP
Vdd	1.7	1.7
Voltage Source	Battery	Battery
Range	Scale1-High	No Scale
Fetch Type	SRAM	n/a
CPU Frequency	100 MHz	0 Hz
Clock Configuration	HSE PLL	Regulator_LPLV Flash-PwrDwn
Clock Source Frequency	4 MHz	0 Hz
Peripherals		
Additional Cons.	0 mA	0 mA
Average Current	12.7 mA	9 µA
Duration	0.1 ms	0.9 ms
DMIPS	125.0	0.0
Ta Max	103.99	105
Category	In DS Table	In DS Table

1.5. Results

Sequence Time	1 ms	Average Current	1.28 mA
Battery Life	3 months, 19 days, 6 hours	Average DMIPS	125.0 DMIPS

1.6. Chart


Consumption Profile by Step
Consumption (mA)
12.5
10.0
7.5
5.0
2.5
0.0
1:RUN
2:STOP
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1.0
Time (ms)
Idd by Step
Average Current

2. Software Project

2.1. Project Settings

Name	Value
Project Name	Hephaestus
Project Folder	C:\Users\toabb\OneDrive\Desktop\RRPL\Hephaestus
Toolchain / IDE	STM32CubeIDE
Firmware Package Name and Version	STM32Cube FW_F4 V1.28.3
Application Structure	Advanced
Generate Under Root	Yes
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

2.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy only the necessary library files
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	No
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

2.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_SPI1_Init	SPI1
4	MX_SPI2_Init	SPI2
5	MX_SPI3_Init	SPI3
6	MX_USB_OTG_FS_PCD_Init	USB_OTG_FS

3. Peripherals and Middlewares Configuration

3.1. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

3.1.1. Parameter Settings:

System Parameters:	
VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	0 WS (1 CPU cycle)
RCC Parameters:	
HSI Calibration Value	16
TIM Prescaler Selection	Disabled
HSE Startup Timout Value (ms)	100
LSE Startup Timout Value (ms)	5000
Power Parameters:	
Power Regulator Voltage Scale	Power Regulator Voltage Scale 1

3.2. SPI1

Mode: Full-Duplex Master

3.2.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	**8.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

3.3. SPI2

Mode: Full-Duplex Master

3.3.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	**8.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

3.4. SPI3

Mode: Full-Duplex Master

3.4.1. Parameter Settings:

Basic Parameters:	
Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	**8.0 MBits/s ***
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

3.5. SYS

Debug: Serial Wire

Timebase Source: SysTick

3.6. USB_OTG_FS

Mode: Device_Only

3.6.1. Parameter Settings:

Speed	Device Full Speed 12MBit/s
Low power	Disabled
Link Power Management	Disabled
VBUS sensing	Disabled
Signal start of frame	Disabled

*** User modified value**

4. System Configuration

4.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
RCC	PH0 - OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1 - OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
SPI1	PA5	SPI1_SCK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	IMU_SCK
	PA6	SPI1_MISO	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	IMU_MISO
	PA7	SPI1_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	IMU_MOSI
SPI2	PC2	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	BAR_MISO
	PC3	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	BAR_MOSI
	PC7	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	BAR_SCK
SPI3	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	MAG/FLASH_SCK
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	MAG/FLASH_MISO
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	MAG/FLASH_MOSI
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	SWDIO
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	SWCLK
USB_OTG_FS	PA11	USB_OTG_FS_DM	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	USB_N
	PA12	USB_OTG_FS_DP	Alternate Function Push Pull	No pull-up and no pull-down	**Very High ***	USB_P
GPIO	PC0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BAR1_CS
	PC1	GPIO_EXTI1	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	BAR1_INT
	PA1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BAR2_CS
	PA2	GPIO_EXTI2	External Interrupt Mode with	No pull-up and no pull-down	n/a	BAR2_INT

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
			Rising edge trigger detection			
	PC4	GPIO_EXTI4	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IMU1_INT
	PC5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IMU1_CS
	PB0	GPIO_EXTI0	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IMU2_GYRO_INT
	PB1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IMU2_GYRO_CS
	PB2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	IMU2_ACC_CS
	PB10	GPIO_EXTI10	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	IMU2_ACC_INT
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED
	PB13	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	BUZZER
	PC8	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	PYRO1_SENSE
	PC9	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	PYRO1
	PA8	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	PYRO2
	PA9	GPIO_Input	Input mode	No pull-up and no pull-down	n/a	PYRO2_SENSE
	PD2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	MAG_CS
	PB3	GPIO_EXTI3	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	MAG_INT
	PB4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	FLASH_CS

4.2. DMA configuration

nothing configured in DMA service

4.3. NVIC configuration

4.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Hard fault interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
System service call via SWI instruction	true	0	0
Debug monitor	true	0	0
Pendable request for system service	true	0	0
System tick timer	true	15	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
RCC global interrupt	unused		
EXTI line0 interrupt	unused		
EXTI line1 interrupt	unused		
EXTI line2 interrupt	unused		
EXTI line3 interrupt	unused		
EXTI line4 interrupt	unused		
SPI1 global interrupt	unused		
SPI2 global interrupt	unused		
EXTI line[15:10] interrupts	unused		
SPI3 global interrupt	unused		
USB On The Go FS global interrupt	unused		
FPU global interrupt	unused		

4.3.2. NVIC Code generation

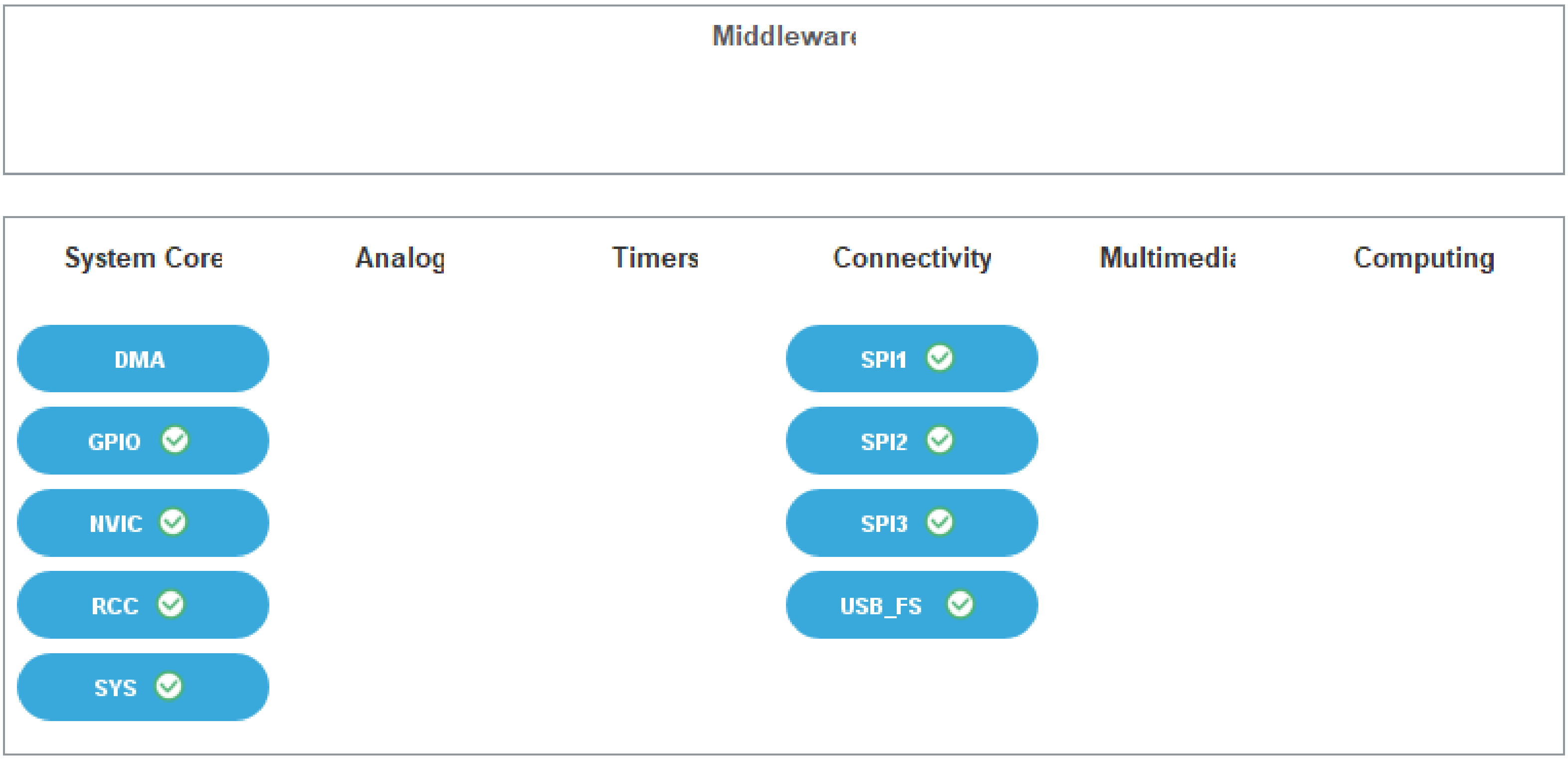
Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	true	false
Hard fault interrupt	false	true	false
Memory management fault	false	true	false
Pre-fetch fault, memory access fault	false	true	false
Undefined instruction or illegal state	false	true	false
System service call via SWI instruction	false	true	false
Debug monitor	false	true	false
Pendable request for system service	false	true	false
System tick timer	false	true	true

*** User modified value**

5. System Views

5.1. Category view

5.1.1. Current



6. Docs & Resources

Type	Link